

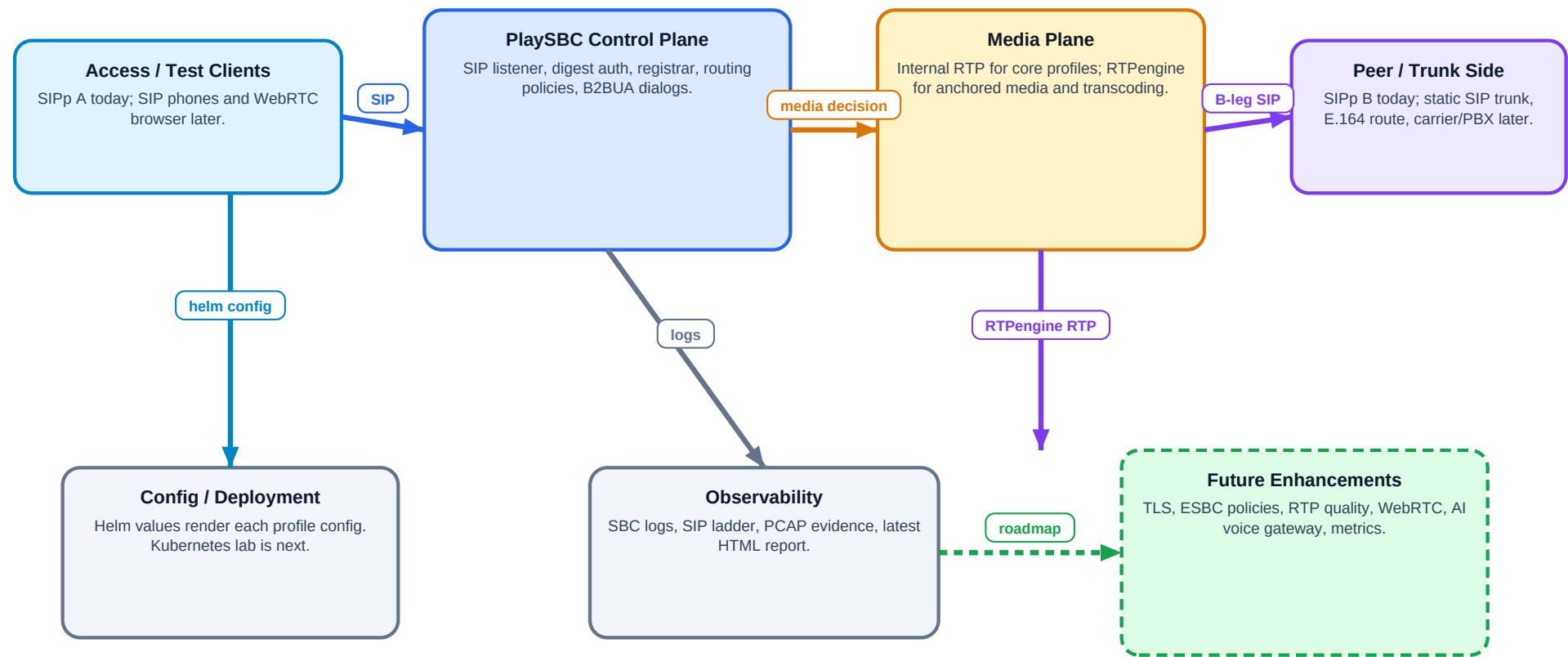
PlaySBC Broad Platform Picture

A lab SBC architecture: access side, SIP control plane, media plane, peer side, config, and evidence.

Direction / responsibility colors

Config SIP control Registrar/routing Internal media RTPengine media Future Logs/report

Break SIP here, not in production. The lawyers will thank the lab.



Core idea: PlaySBC owns signalling, routing, dialog state, and evidence. RTPengine is the optional external media backend.

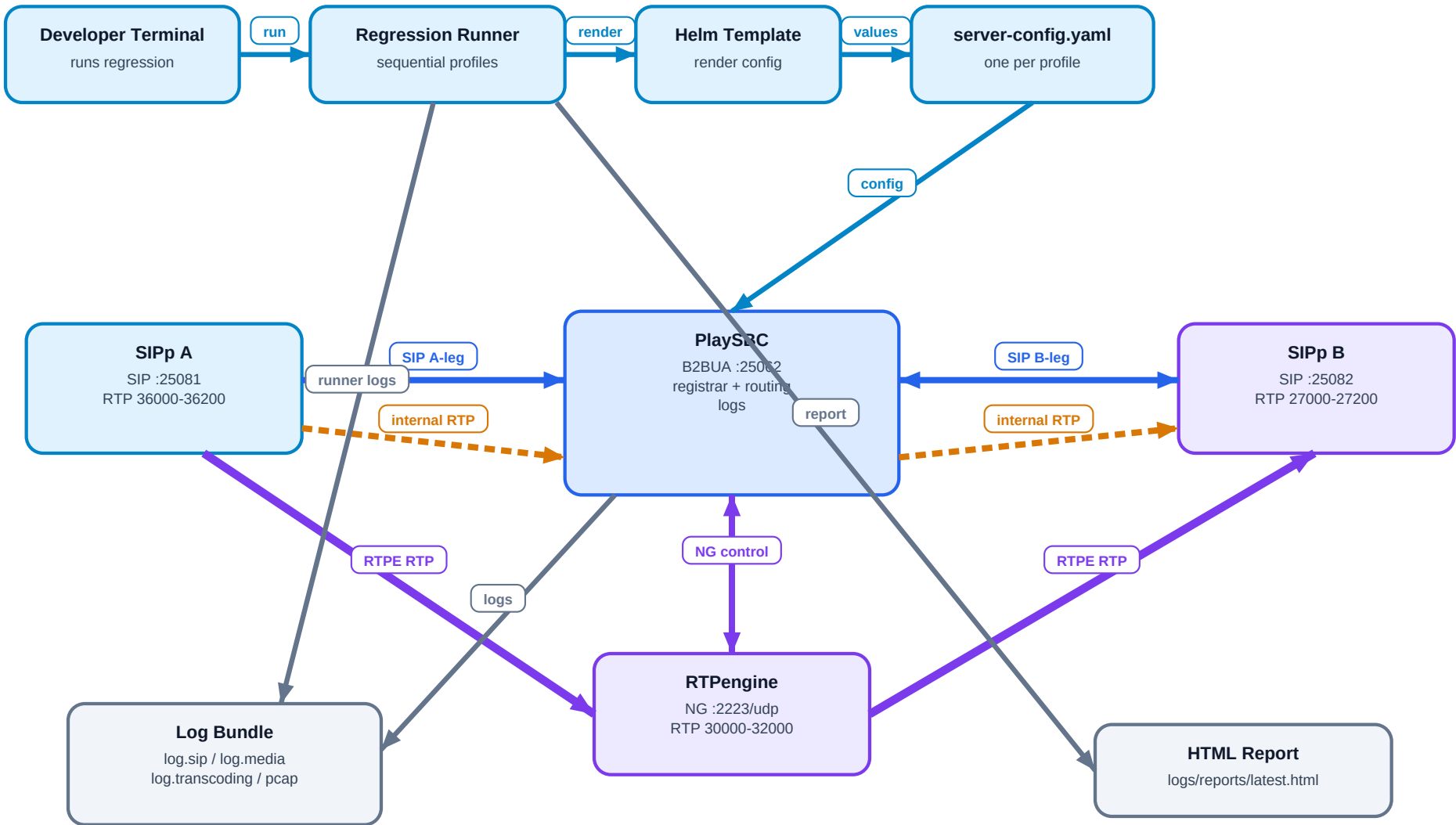
Current Local Service Network

Default local ports and direction colors used by SIPp regression.

Loopback IPs, enterprise habits. Tiny lab, serious protocol receipts.

Direction / responsibility colors

- Config
- SIP control
- Registrar/routing
- Internal media
- RTPengine media
- Future
- Logs/report

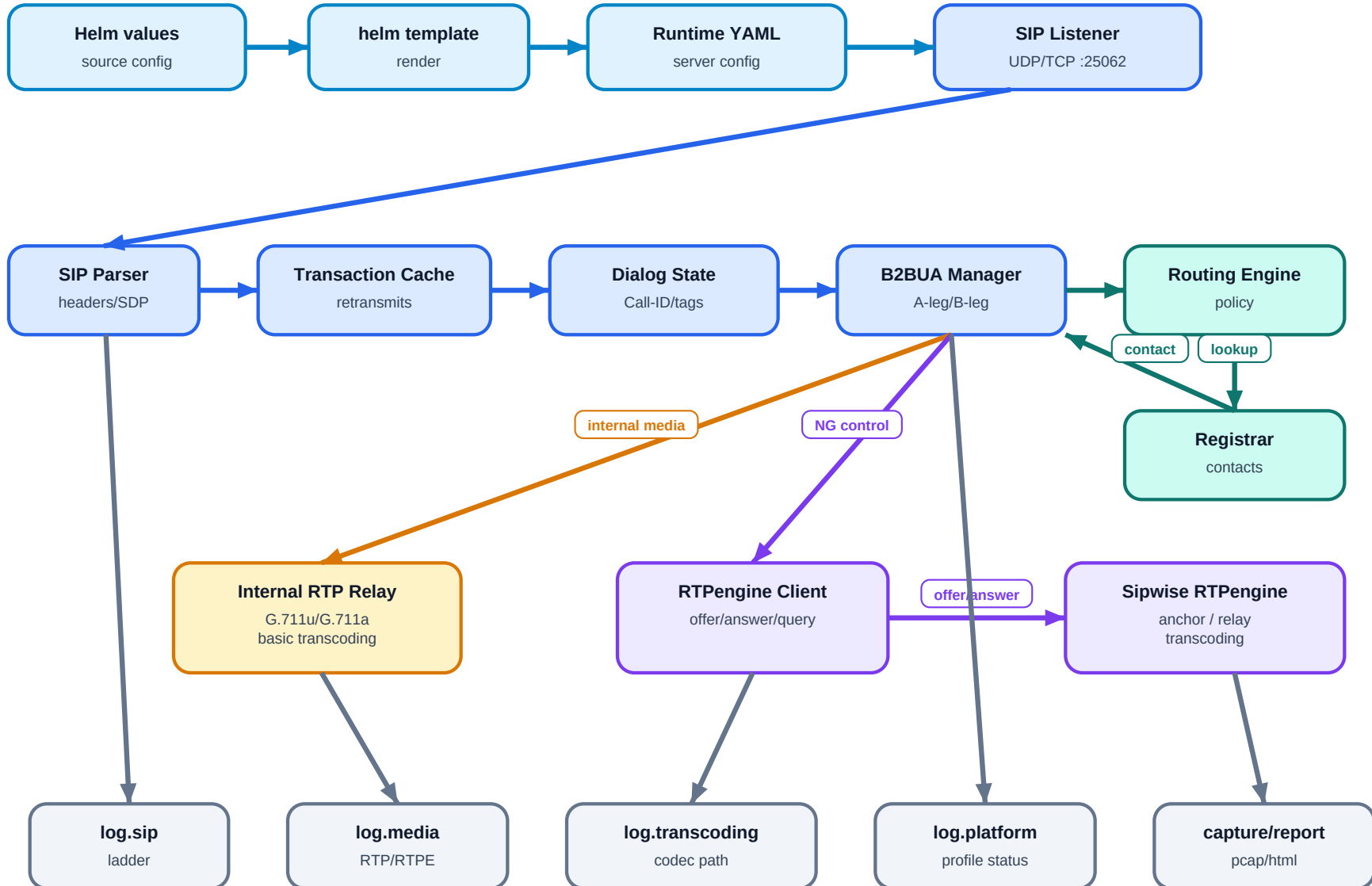


PlaySBC Low-Level Components

Internal responsibilities: parse, authenticate, route, create B-leg, choose media backend, log evidence.

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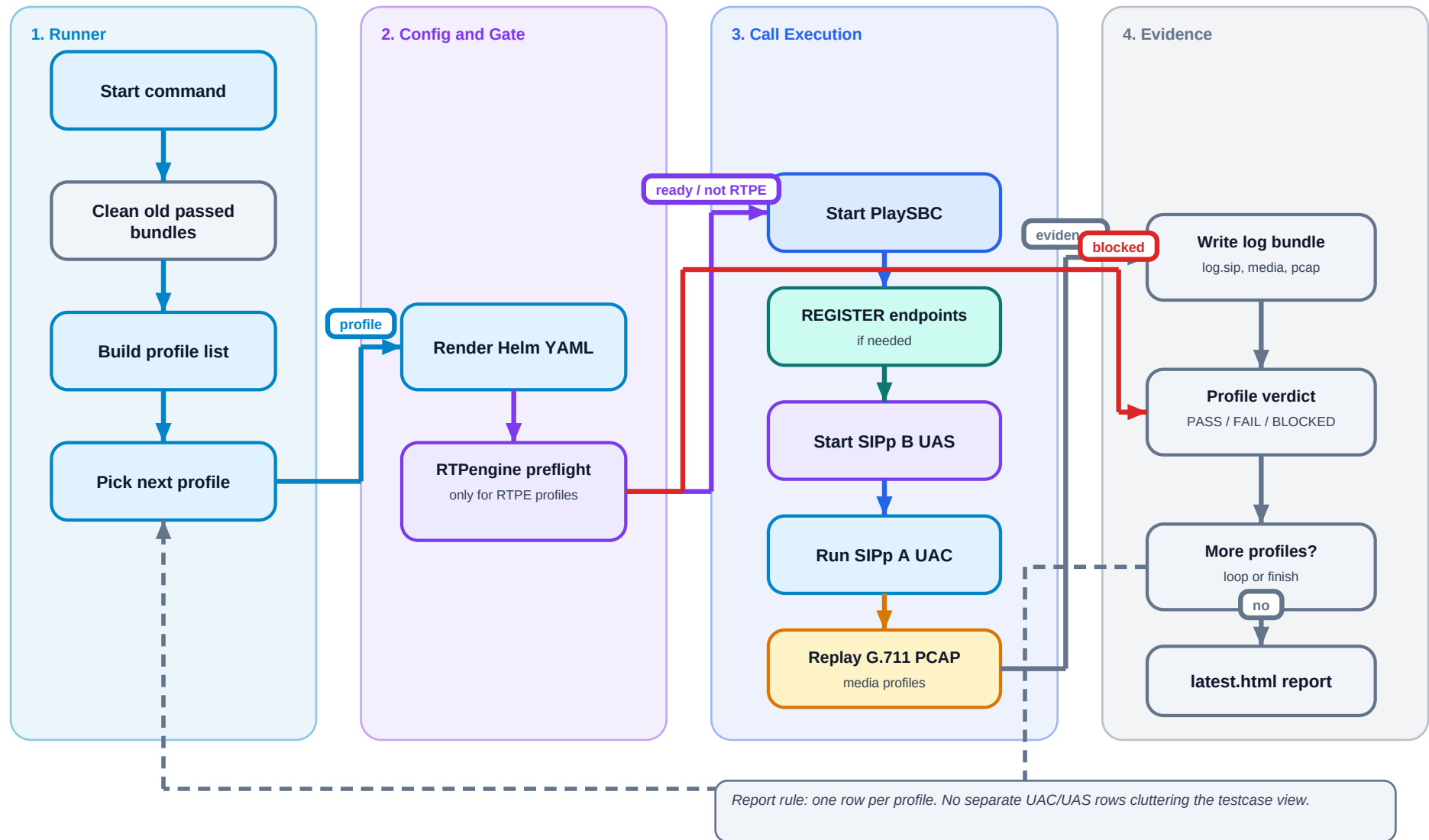
SIPp Regression Testing Flow

Clean swimlanes, one profile at a time. Load profiles create concurrency inside the profile only.

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If RTPengine is asleep, the profile is marked **BLOCKED**. No fake failures, no mystery-meat reports.

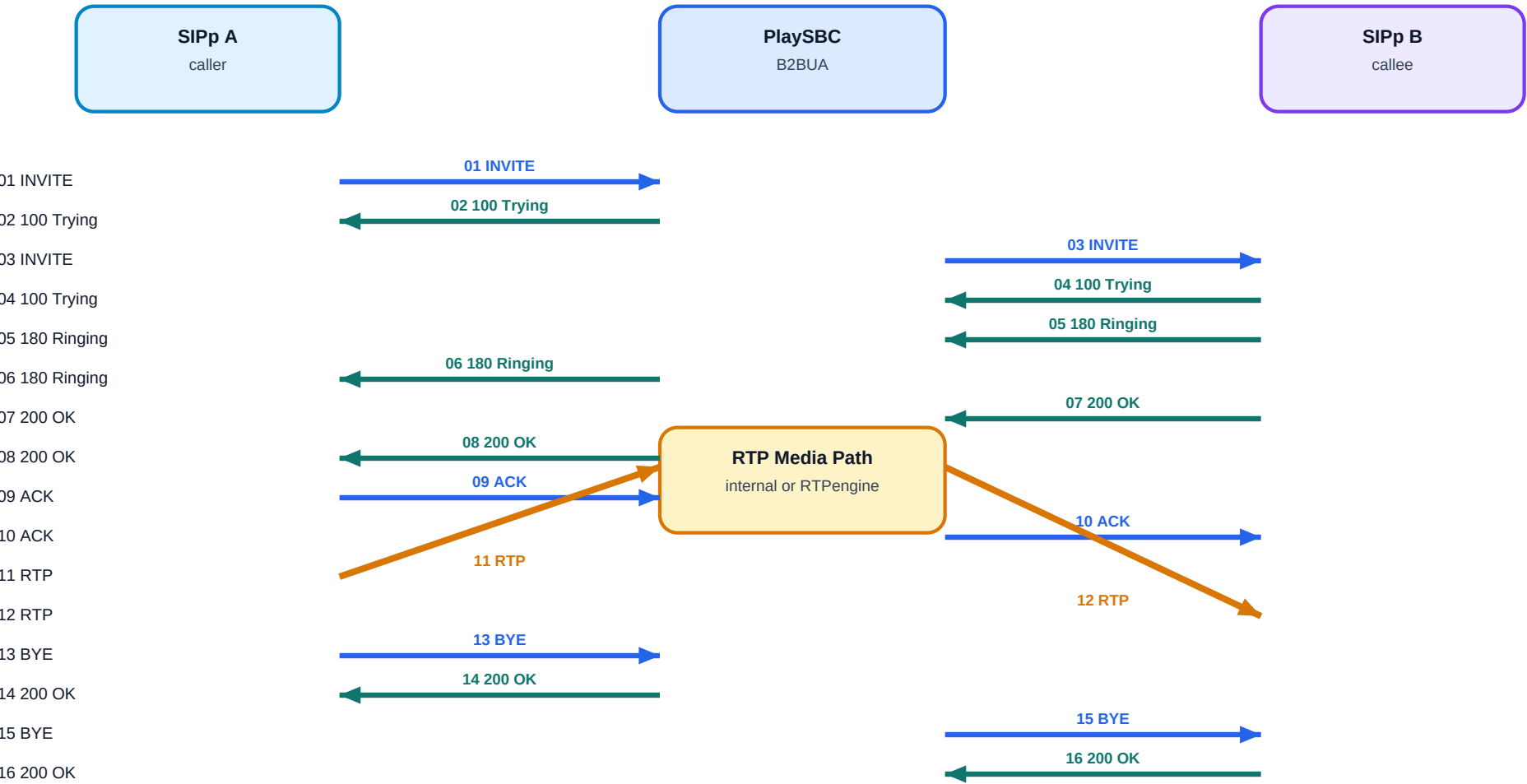


Basic B2BUA Call Path

Technical flow for one call. SIP requests/responses remain anchored in PlaySBC.

Direction / responsibility colors

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In RTPengine profiles, PlaySBC controls media by NG offer/answer/query, but RTP packets bypass PlaySBC and traverse RTPengine.

Logical Node Glossary - Service Nodes

Each logical node, what it means, and one concrete example.

Logical node	Technical purpose	Example
Developer Terminal	Local shell used to start readiness checks and full regression.	<code>python3 tools/run_regression_suite.py --all-b2bua-profiles</code>
Regression Runner	Sequential orchestrator. Renders config, starts services, captures logs, records result.	basic-media runs, stops, then transcoding starts.
Helm Template	Config renderer used locally via helm template. Kubernetes install is not required for regression.	<code>helm template playsbc charts/playsbc</code>
Per-profile Config	Temporary YAML for one test profile only. Keeps TCP/RTPengine/load settings isolated.	<code>sip_transport: tcp, media_backend: rtpengine</code>
SIPp A	Caller side: UAC or registered caller. Sends inbound INVITE to PlaySBC.	<code>INVITE sip:callee@127.0.0.1:25062 SIP/2.0</code>
PlaySBC	SIP registrar, router, transaction/dialog owner, B2BUA leg creator, and log owner.	A-leg INVITE becomes a new B-leg INVITE.
SIPp B	Callee side: UAS or registered endpoint. Responds to B-leg from PlaySBC.	100 Trying, 180 Ringing, 200 OK with SDP
RTPengine	Optional media backend used for anchoring and transcoding experiments.	NG: <code>udp://127.0.0.1:2223</code> , RTP pool 30000-32000

Logical Node Glossary - PlaySBC Internals

Core internal responsibilities inside the PlaySBC process.

Internal node	Technical purpose	Example
SIP Listener	UDP/TCP socket entry point for SIP messages on local regression port 25062.	REGISTER, OPTIONS, INVITE, ACK, CANCEL, BYE
SIP Parser	Extracts method, headers, Call-ID, CSeq, tags, Contact, and SDP body.	Find Contact in REGISTER and m=audio in SDP.
Transaction Cache	Handles retransmitted SIP requests consistently.	Duplicate INVITE branch/CSeq uses cached behavior.
Dialog State	Tracks call identity across A-leg and B-leg.	Call-ID, From tag, To tag, route target.
Registrar	In-memory location service for registered users.	callee -> sip:callee@127.0.0.1:25082
Routing Engine	Chooses registrar, static trunk, E.164 policy, or rejection.	+1800* -> sip:{user}@127.0.0.1:25080
B2BUA Manager	Creates and coordinates outbound B-leg from inbound A-leg.	Receives INVITE from A, sends INVITE to B.
Internal RTP Relay	PlaySBC-owned media path for core profiles.	G.711u/G.711a relay and basic transcoding.
RTPengine Client	NG protocol client for external media anchoring.	offer, answer, query, delete
Log Bundle	One testcase folder with signalling, media, transcoding, platform, PCAP/report evidence.	log.sip, log.media, log.transcoding, capture.pcap

Configuration Examples

Small YAML examples that map directly to the network diagrams.

Core B2BUA profile - internal media

```
sip_ip: 127.0.0.1
sip_port: 25062
sip_transport: udp
media_backend: internal
default_codec: PCMU
route_policies:
  - name: registered-endpoints
    match: "*"
    target: registration
    priority: 10
```

RTPengine profile - external media backend

```
sip_ip: 127.0.0.1
sip_port: 25062
sip_transport: udp
media_backend: rtpengine
rtpengine_url: udp://127.0.0.1:2223
rtpengine_timeout: 3.0
b2bua_ladder_logs: true
reject_unknown_routes: false
```

ESBC routing policy example

```
users:
  enterprise-a: enterprise-a-pass
route_policies:
  - name: esbc-e164-outbound
    match: "+1800*"
    target: "sip:{user}@127.0.0.1:25080"
    priority: 20
  - name: esbc-static-trunk
    match: "trunk-*"
    target: "sip:{user}@127.0.0.1:25080"
    priority: 30
reject_unknown_routes: true
```

How to read this

Core profiles keep SIP and RTP in the lab with PlaySBC internal media. RTPengine profiles keep SIP in PlaySBC but anchor RTP in RTPengine. ESBC policies choose the outbound leg using registrar, static trunk, or E.164 matching.

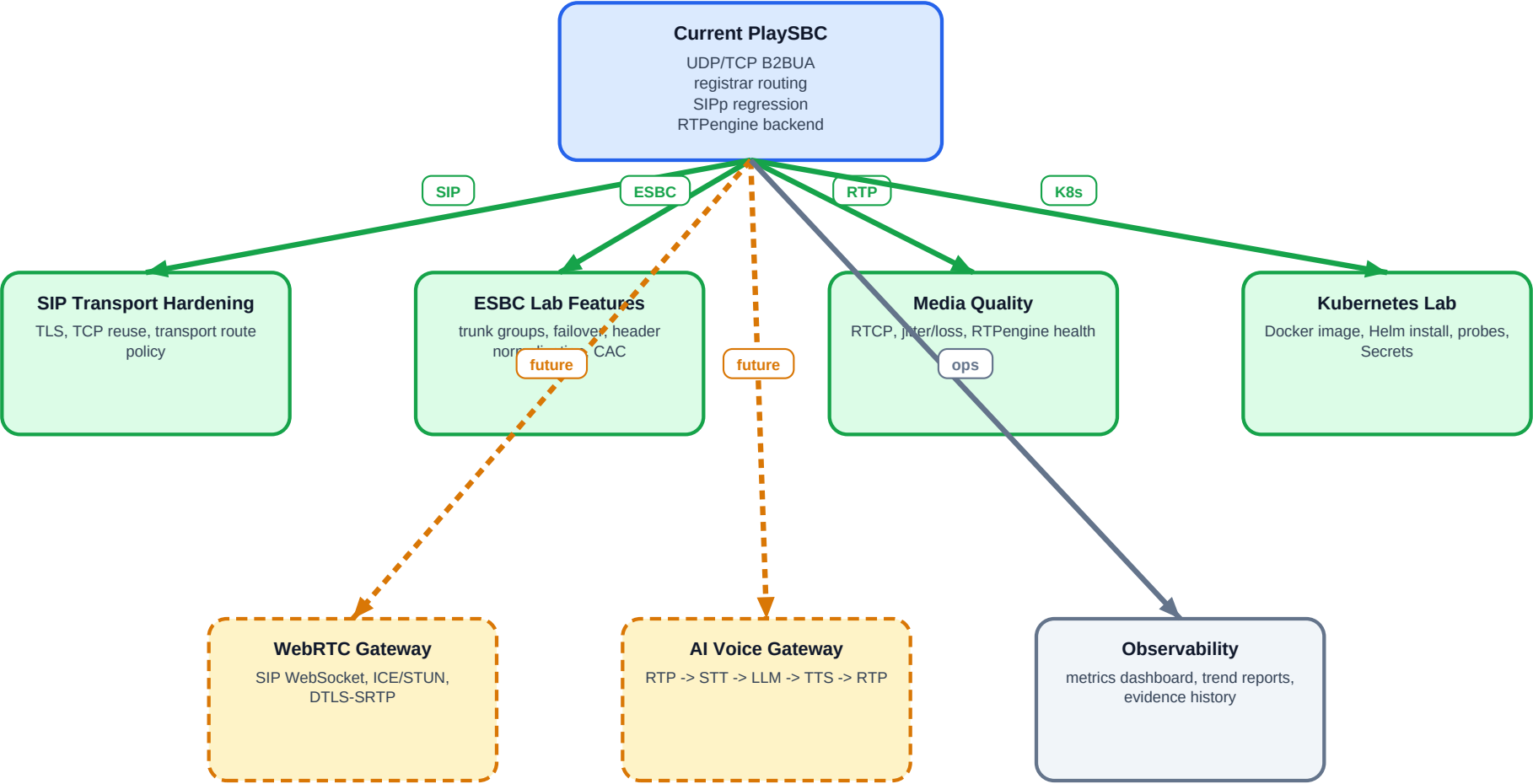
Future Enhancement Roadmap View

Broader target architecture beyond the current local SIPp/RTPengine lab.

Roadmap rule: if it cannot be tested, logged, and explained, it is just a wish with YAML.

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Guiding rule: each enhancement should add config, SIPp or unit coverage, SBC-style logs, and clear pass/fail evidence.